

Lab Sheet 8: ES6 Basics

Branch/ Class: B.Tech/M.Tech

Date:

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name:

Reg. no.:

Exercise 1

Write a JavaScript program using ES6 features to calculate the total and average marks of a student. The program should declare variables using `let` and `const`, use an arrow function to compute the average, and display the result using template literals. The program should print the student name, total marks, and average marks in a formatted message.

Sample Code Structure

```
let studentName = "Arun";
let mark1 = 85;
let mark2 = 90;
let mark3 = 88;
const calculateAverage = (m1, m2, m3) => {
  return (m1 + m2 + m3) / 3;
}
let average = calculateAverage(mark1, mark2, mark3);
console.log(`Student Name: ${studentName}`);
console.log(`Average Marks: ${average}`);
```

```
js > ...
let studentName = "Arun";
let mark1 = 85;
let mark2 = 90;
let mark3 = 88;

let total = mark1 + mark2 + mark3;

const calculateAverage = (m1, m2, m3) => {
  return ((m1 + m2 + m3) / 3).toFixed(2);
};

let average = calculateAverage(mark1, mark2, mark3);

console.log(`Student Name: ${studentName}`);
console.log(`Total Marks: ${total}`);
console.log(`Average Marks: ${average}`);
```

Expected Output

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Student Name: Arun

Average Marks: 87.66

```
C:\Users\patna\OneDrive\Desktop\WebTech\lab8>node Ex1.js
Student Name: Arun
Total Marks: 263
Average Marks: 87.67
```

Exercise 2

Create a JavaScript program that defines an object representing a student with properties such as id, name, department, and marks. Use object destructuring to extract the values and display them. Then create a new object using the spread operator that adds an additional property called grade based on the marks and display the updated object.

Sample Code Structure

```
const student = {
  id: 101,
  name: "Priya",
  department: "CSE",
  marks: 92
};

const {id, name, department, marks} = student;
console.log(id, name, department, marks);
const updatedStudent = {
  ...student,
  grade: "A"
};
console.log(updatedStudent);
```

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```
const student = {
  id: 101,
  name: "Priya",
  department: "CSE",
  marks: 92
};

const {id, name, department, marks} = student;

console.log(id, name, department, marks);

const updatedStudent = {
  ...student,
  grade: "A"
};

console.log(updatedStudent);
```

Expected Output

101 Priya CSE 92

{ id: 101, name: 'Priya', department: 'CSE', marks: 92, grade: 'A' }

```
C:\Users\patna\OneDrive\Desktop\WebTech\lab8>node Ex2.js
```

```
101 Priya CSE 92
```

```
{ id: 101, name: 'Priya', department: 'CSE', marks: 92, grade: 'A' }
```

Exercise 3

Develop a JavaScript program using ES6 classes and promises to simulate a simple course enrollment system. Create a class named Course with a constructor to initialize the course name and instructor. Include a method to display course details. Implement a promise that checks whether seats are available and prints either "Enrollment Successful" or "Course Full" based on the condition.

Sample Code Structure

```
class Course {
  constructor(courseName, instructor) {
    this.courseName = courseName;
```

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```
        this.instructor = instructor;
    }
    displayCourse() {
        console.log(` Course: ${this.courseName}, Instructor: ${this.instructor}`);
    }
}
let course1 = new Course("Web Technologies", "Dr. Kumar");
course1.displayCourse();
let enrollCourse = new Promise((resolve, reject) => {
    let seatsAvailable = true;
    if(seatsAvailable)
        resolve("Enrollment Successful");
    else
        reject("Course Full");
});
enrollCourse
    .then(msg => console.log(msg))
    .catch(err => console.log(err));
```

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```
class Course {
  constructor(courseName, instructor) {
    this.courseName = courseName;
    this.instructor = instructor;
  }

  displayCourse() {
    console.log(`Course: ${this.courseName}, Instructor: ${this.instructor}`);
  }
}

let course1 = new Course("Web Technologies", "Dr. Kumar");

course1.displayCourse();

let enrollCourse = new Promise((resolve, reject) => {
  let seatsAvailable = true;

  if(seatsAvailable)
    resolve("Enrollment Successful");
  else
    reject("Course Full");
});

enrollCourse
  .then(msg => console.log(msg))
  .catch(err => console.log(err));
```

Expected Output

Course: Web Technologies, Instructor: Dr. Kumar

Enrollment Successful

```
C:\Users\patna\OneDrive\Desktop\WebTech\lab8>node Ex3.js
Course: Web Technologies, Instructor: Dr. Kumar
Enrollment Successful
```